

4.3.2 Internal energy and energy transfers

4.3.2.1 Internal energy

Content	Key opportunities for skills development
Energy is stored inside a system by the particles (atoms and molecules) that make up the system. This is called internal energy. Internal energy is the total kinetic energy and potential energy of all the particles (atoms and molecules) that make up a system. Heating changes the energy stored within the system by increasing the energy of the particles that make up the system. This either raises the temperature of the system or produces a change of state.	

4.3.2.2 Temperature changes in a system and specific heat capacity

Content	Key opportunities for skills development
If the temperature of the system increases, the increase in temperature depends on the mass of the substance heated, the type of material and the energy input to the system.	
The following equation applies: <i>change in thermal energy = mass × specific heat capacity × temperature change</i> $[\Delta E = m c \Delta \theta]$ change in thermal energy, ΔE, in joules, J mass, m, in kilograms, kg specific heat capacity, c, in joules per kilogram per degree Celsius, J/kg °C temperature change, $\Delta \theta$, in degrees Celsius, °C. The specific heat capacity of a substance is the amount of energy required to raise the temperature of one kilogram of the substance by one degree Celsius.	MS 1a, 3b, c, d Students should be able to apply this equation, which is given on the Physics equation sheet, to calculate the energy change involved when the temperature of a material changes. This equation and specific heat capacity are also included in Energy changes in systems (page 19).

4.3.2.3 Changes of state and specific latent heat

Content	Key opportunities for skills development
If a change of state happens: The energy needed for a substance to change state is called latent heat. When a change of state occurs, the energy supplied changes the energy stored (internal energy) but not the temperature. The specific latent heat of a substance is the amount of energy required to change the state of one kilogram of the substance with no change in temperature.	

Content	Key opportunities for skills development
<i>energy for a change of state = mass × specific latent heat</i> [$E = m L$] energy, E, in joules, J mass, m, in kilograms, kg specific latent heat, L, in joules per kilogram, J/kg Specific latent heat of fusion – change of state from solid to liquid Specific latent heat of vaporisation – change of state from liquid to vapour	MS 1a, 3b, c, d Students should be able to apply this equation, which is given on the <i>Physics equation sheet</i>, to calculate the energy change involved in a change of state. MS 4a AT 5 Perform an experiment to measure the latent heat of fusion of water.
Students should be able to interpret heating and cooling graphs that include changes of state. Students should be able to distinguish between specific heat capacity and specific latent heat.	WS 3.5